

Improper integrals – 2

Conditional convergence and oscillation

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Theoretical minimum.

Theorem 1 (Dirichlet and Abel). Let f be monotone with $f(x) \rightarrow 0$ and let $G(b) = \int_a^b g$ be bounded; then $\int_a^\infty fg$ converges. Let f be monotone and bounded and let $\int_a^\infty g$ converge; then $\int_a^\infty fg$ converges. Both follow from the second mean value theorem in Bonnet's form, proved in Definite integrals – 6: $\int_{b_1}^{b_2} fg = f(b_1) \int_{b_1}^\xi g$ for non-negative non-increasing f .

Theorem 2 (the phase principle). If φ is C^1 with φ' monotone and $\varphi'(x) \rightarrow \infty$, then $\int^\infty \sin \varphi(x) dx$ converges: after the substitution $u = \varphi(x)$ the integrand becomes $\frac{\sin u}{\varphi'(\varphi^{-1}(u))}$ and Dirichlet's test applies. The integrand itself need not tend to zero — $\int_1^\infty \sin(x^2) dx$ is the standard example.

Theorem 3 (Riemann–Lebesgue). For integrable f on $[a, b]$, $\int_a^b f(x) \sin(\lambda x) dx \rightarrow 0$ as $\lambda \rightarrow \infty$; for monotone f the rate is $O(1/\lambda)$ by Bonnet, and in general no rate is possible. On an infinite interval the statement needs absolute convergence of $\int |f|$.

Fact 1 (absolute versus conditional). $\int_1^\infty \frac{\sin x}{x} dx$ converges while $\int_1^\infty \frac{|\sin x|}{x} dx = \infty$, because the mean value of $|\sin|$ over a period is $\frac{2}{\pi}$ and hence $\int_1^T \frac{|\sin x|}{x} dx = \frac{2}{\pi} \ln T + O(1)$. The same computation is the model for every proof of non-absolute convergence.

Fact 2 (integration by parts is the engine). Writing $\int f \sin x dx = [-f \cos x] + \int f' \cos x dx$ converts an oscillatory integral into an absolutely convergent one as soon as f' is integrable and $f \rightarrow 0$; this is how one both proves convergence and obtains asymptotics of the tail.

1. Prove the tests of Dirichlet and Abel for improper integrals from Bonnet's theorem. Deduce that $\int_1^\infty \frac{\sin x}{x^p} dx$ converges for $p > 0$ and converges absolutely exactly for $p > 1$, and prove the asymptotics $\int_1^T \frac{|\sin x|}{x} dx = \frac{2}{\pi} \ln T + O(1)$.
2. Prove that $\int_1^\infty \sin(x^2) dx$ converges, that the integrand does not tend to zero and is not even bounded away from 1 on a set of infinite measure, and prove Theorem 2 in general. (The value of the Fresnel integral is computed by contour integration in Complex analysis – 4.)
3. (VJIMC 2006) Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a strictly convex continuous function with $\lim_{x \rightarrow +\infty} \frac{f(x)}{x} = +\infty$. Prove that the improper integral $\int_0^{+\infty} \sin(f(x)) dx$ is convergent but not absolutely convergent.
4. (S.T. Yau CSMC 2024) Let $\varphi \in C_c^\infty(\mathbb{R})$ be a compactly supported smooth function.
 - a) Prove that $\lim_{\lambda \rightarrow +\infty} \int_{-\infty}^{+\infty} e^{i\lambda x^2} \varphi(x) dx = 0$.
 - b) Prove that there are constants $\alpha > 0$ and $C > 0$ such that $|\int_{-\infty}^{+\infty} e^{i\lambda x^2} \varphi(x) dx| \leq C\lambda^{-\alpha}$ for all $\lambda \geq 1$.
5. Prove the Riemann–Lebesgue lemma for a Riemann integrable f on $[a, b]$, show that for monotone f the rate $O(1/\lambda)$ is sharp, and prove that no rate is valid for all continuous functions: for every $\varepsilon(\lambda) \rightarrow 0$ there is a continuous f with $|\int_0^1 f(x) \sin(\lambda x) dx| \geq \varepsilon(\lambda)$ for arbitrarily large λ .